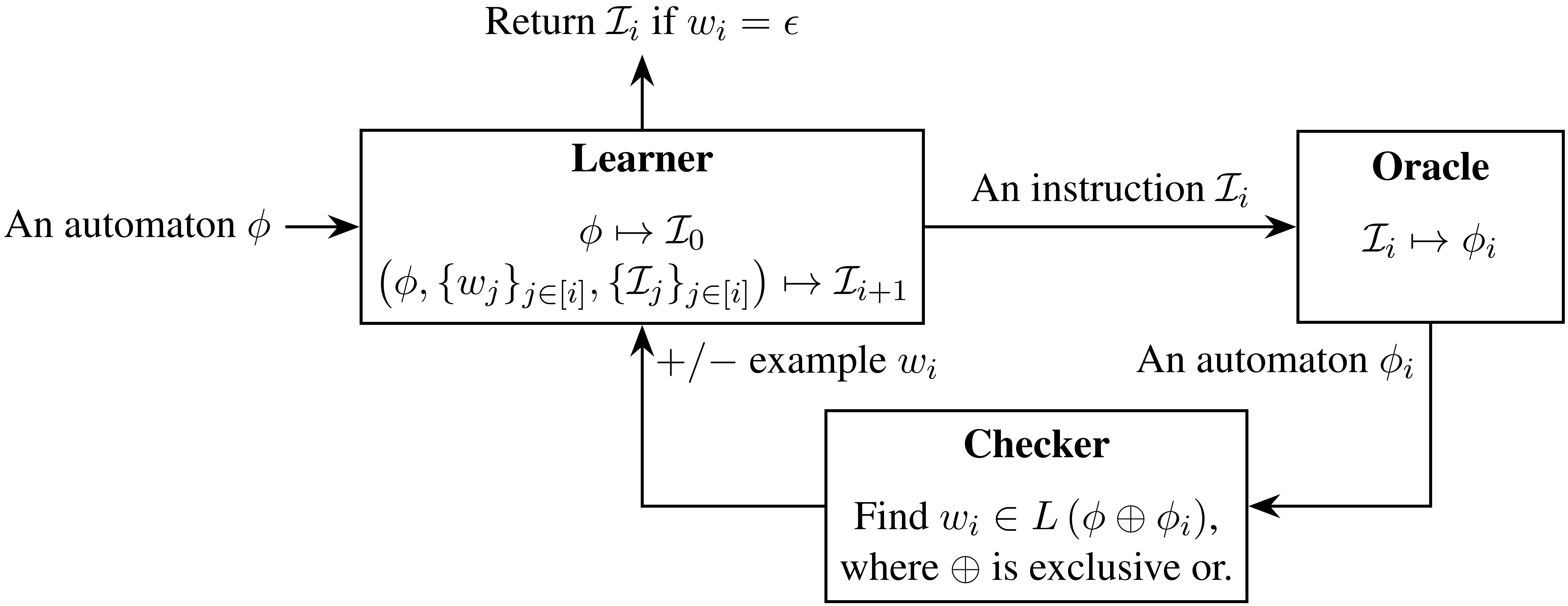
Synthesizing Language Instructions from Automata

- Our approach combines **Counterexample-Guided Inductive Synthesis (CEGIS)** with **Large Language Models (LLMs)**.

- **Assumption:** An instruction is said to be consistent with an automaton if an LLM can zero-shot translate it to an automaton with the same language.



Return $\mathcal{I}_i$ if $w_i = \epsilon$
Learner
$\phi \mapsto \mathcal{I}_0$
$\left(\phi, \{w_j\}_{j\in[i]}, \{\mathcal{I}_j\}_{j\in[i]}\right) \mapsto \mathcal{I}_{i+1}$
An automaton ϕ
An instruction $\mathcal{I}_i$
Oracle
$\mathcal{I}_i \mapsto \phi_i$
An automaton ϕ_i
$+/-$ example w_i
Checker
Find $w_i \in L\left(\phi \oplus \phi_i\right)$,
where $\oplus$ is exclusive or.

Solar-Lezama, Armando, et al. "Combinatorial sketching for finite programs." ASPLOS 2006.
Brown, Tom, et al. "Language models are few-shot learners." Neurips 2020.

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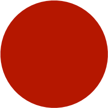

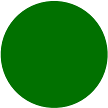

Neural (LLM)

Symbolic

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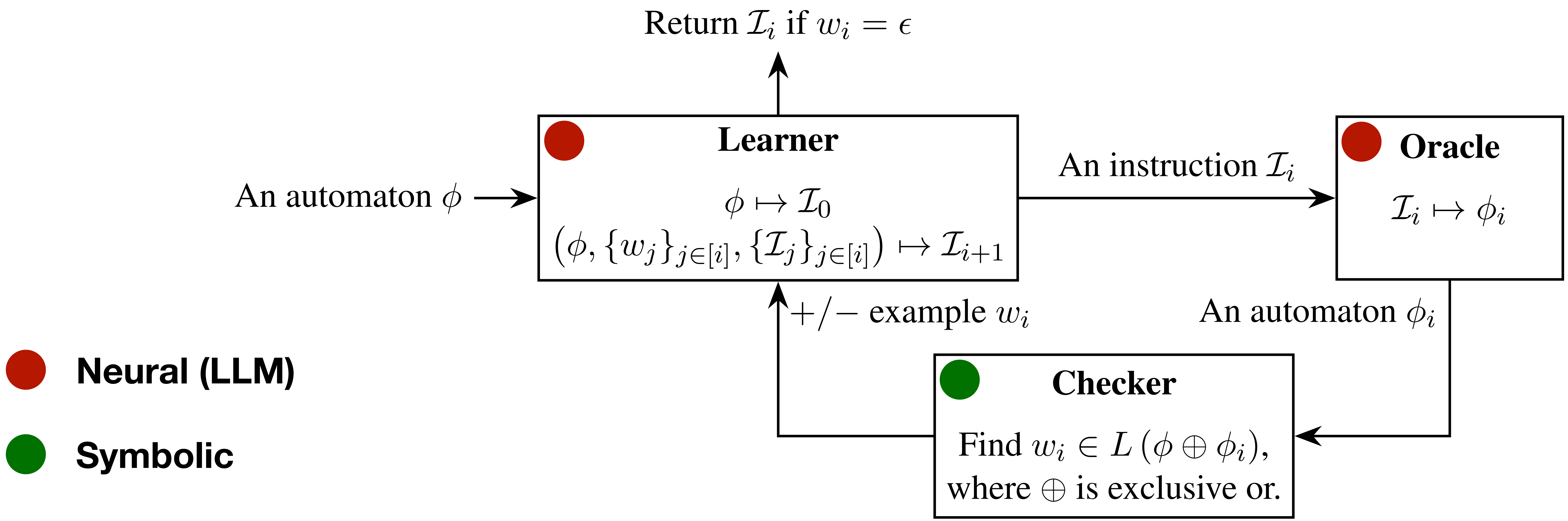

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Outline

- ~~Introduction~~
- ~~Preliminaries~~
- ~~Synthesizing Language Instructions from Automata~~
- **Learning Policies with Automata and Instruction Embeddings**
- Empirical Study
- Discussion